

Cancer Nanotechnology User Guide

The purpose of this document is to provide guidance on cancer nanotechnology data available in the General Commons (GC). It is organized into two parts to help users interact with the GC for both data discovery and data submission through the CRDC Data Submission Portal.

Section 1: Searching Data in General Commons (GC) - explains how to search the GC portal and access cancer nanotechnology data.

Section 2: Submitting Data in General Commons (GC) - provides instructions on how to submit cancer nanotechnology data to the CRDC for inclusion in the GC.

Section 1: Searching Data in General Commons (GC)

I. Filter and Search for Nanotechnology Studies

1. Go to <https://general-ga2.datacommons.cancer.gov/>
2. On the homepage, click **Data** to open the Data page where studies can be searched using filters (see Figure 1).

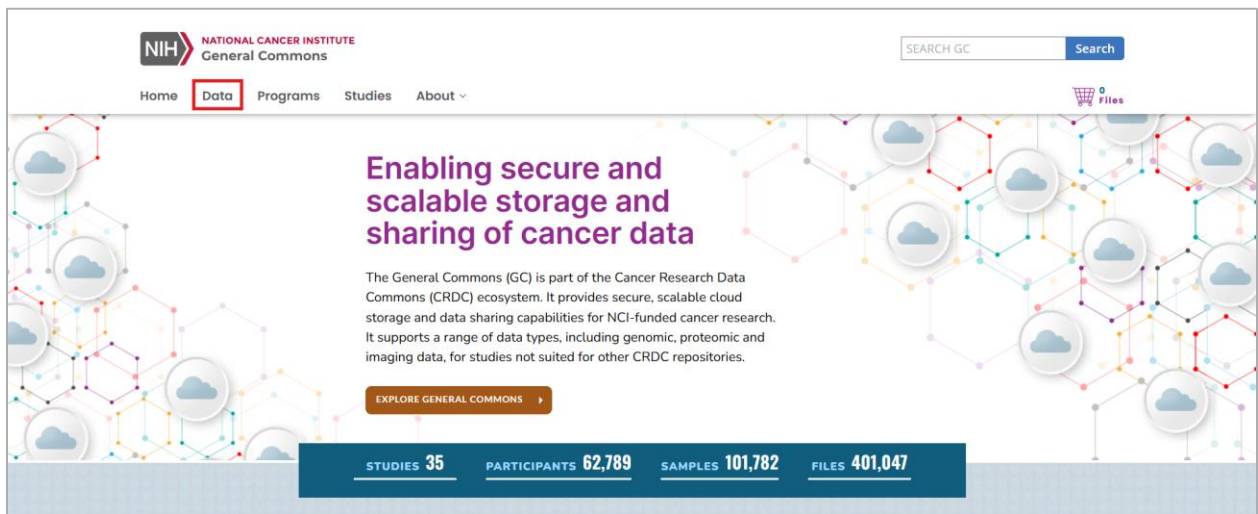


Figure 1. General Commons homepage with the “Data” tab highlighted to access study filters.

3. On the left navigation bar of the Data Page, click the black triangle next to **STUDY DATA TYPES** to expand the menu (see Figure 2).
4. Check the box for **Nanotechnology** (see Figure 2).

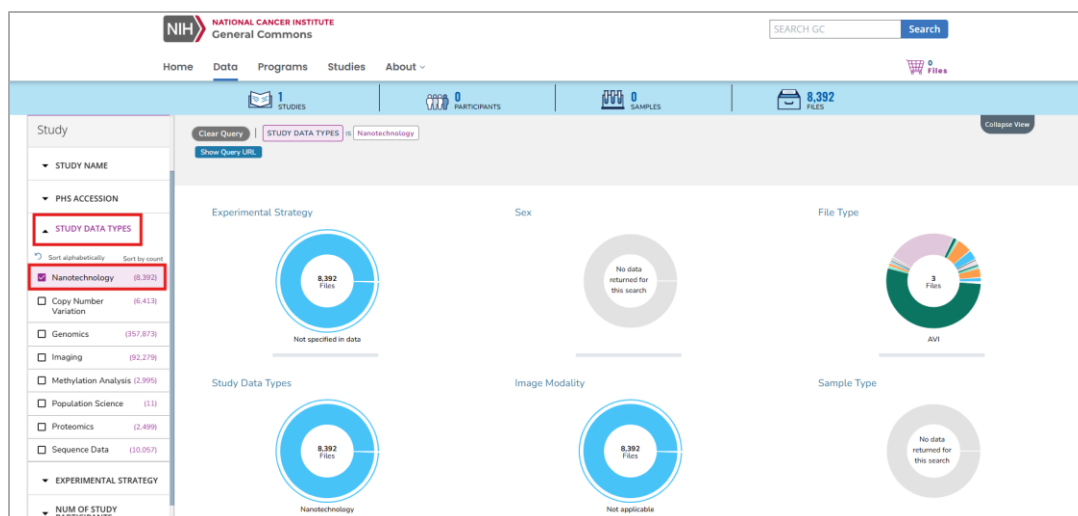


Figure 2. Expanding the STUDY DATA TYPES filter to select Nanotechnology.

5. Scroll down to the **Nanotechnology** section at the bottom of the left navigation bar. You can further refine your search using the following 8 filters:
 - PROTOCOL NAME
 - PROTOCOL TYPE
 - PUBLICATION TITLE
 - NANOMATERIAL ENTITY
 - FUNCTIONALIZING ENTITY
 - CHARACTERIZATION TYPE
 - CHARACTERIZATION NAME
6. Click the black triangle next to any filter to expand its options.
7. Click an option to apply the filter. Matching studies will appear in the lower-right portion of the page (see Figure 3).

Figure 3. Expanded filters and matching study results displayed on the Data page.

II. Access and Export Files

1. If your search returns associated Samples or Files, a number will appear next to each tab (e.g., Samples (1), Files (2)). Click **Files** (or **Samples**) tab to view the corresponding records (See Figure 4).
2. Select file(s) by:
 - Clicking the square next to each file, or
 - Clicking **Select All** to choose all listed files (see Figure 4).
3. Click **Add Selected Files** or **Add All Files** (see Figure 4) to add the selected files to the cart.

Cancer Nanotechnology	Participants (0)		Samples (0)		Files (2)	
					ADD ALL FILES ADD SELECTED FILES	
PROTOCOL NAME Sort alphabetically Sort by count ☒ CD34+ CD90+ Digital Magnetic Cell Sorting Assay (1) ☒ CellTiter 96 AQueous One Solution Cell Proliferation Assay (1) ☐ CPMV-S100A9 Agarose Gel Electrophoresis Protocol (1) ☐ CPMV-S100A9 DLS Protocol (1) ☐ CPMV-S100A9 Lung Metastasis Model Protocol (1)	10 row(s) selected				ROWS PER PAGE: 10 1-2 OF 2	
	Study Name	Accession	File Name ↑	File ID	File Type	Sample ID
☒	caNanoLab Data	GC000001	20061211_10-30-46-773_NCL200612A_fig 20.jpg	dg.4DFC/47d5047a-5bff-5469-9c3d-0fb39614dd13	JPG	Not applicable
☒	caNanoLab Data	GC000001	20061211_11-06-47-966_NCL200612A_fig 14.jpg	dg.4DFC/d28ae41d-00cb-5411-8a27-470736692a91	JPG	Not applicable
☒	caNanoLab Data	GC000001	20061211_11-29-07-542_NCL200612A_fig 27.jpg	dg.4DFC/0a766ef7-7b14-5bec-90cd-d933b87f579c	JPG	Not applicable
☒	caNanoLab Data	GC000001	20061211_12-47-57-977_NCL200612A_fig 22.jpg	dg.4DFC/6ee688d2-9a90-52da-8dfa-2ece580094a9	JPG	Not applicable
☒	caNanoLab Data	GC000001	20061211_14-49-01-102_NCL200612A_fig 4.jpg	dg.4DFC/86328a1d-b433-52ee-bb23-4f390765a5dd	JPG	Not applicable
☒	caNanoLab Data	GC000001	20061211_14-54-29-189_NCL200612A_fig 5.jpg	dg.4DFC/ec6c54cd-94c9-5ae8-9a49-ec908663b9c4	JPG	Not applicable
☒	caNanoLab Data	GC000001	20061211_15-15-15-306_NCL200612A_fig 6.jpg	dg.4DFC/b12d0a79-6247-5f18-a682-e4dc0ac6b674	JPG	Not applicable
☒	caNanoLab Data	GC000001	20061211_15-25-24-13_NCL200612A_fig 10.jpg	dg.4DFC/e2fc2c13-6d0e-5bf7-bab8-4473de510466	JPG	Not applicable
☒	caNanoLab Data	GC000001	20061211_15-47-58-562_NCL200612A_fig 14.jpg	dg.4DFC/bb916795-da7b-59fa-9c49-67d83e6973ae	JPG	Not applicable
☒	caNanoLab Data	GC000001	20061211_16-03-11-419_NCL200612A_fig 24.jpg	dg.4DFC/f0104b90-e6e1-541b-bd88-9ff5423ffcca	JPG	Not applicable

Figure 4. File view showing checkboxes for selecting individual or all files.

4. Your selected files will appear in the **Cart** (top right of the page) (see Figure 5).

NATIONAL CANCER INSTITUTE General Commons		SEARCH GC Search	
Home	Data	Programs	Studies About ▾

Figure 5. Cart icon at the top right shows the number of selected files added.

5. Click the **Cart icon** to view and export files (see Figure 6).

Cart > Selected Files

README ?

AVAILABLE EXPORT OPTIONS

DOWNLOAD MANIFEST ?

File Name ↑	Study Name	Accession	Participant Id	Sample Id	Study Access	File Type	Remove ▾
20061211_10-30-46-773_NCL200612A_fig 20.jpg	caNanoLab Data	GC000001		Not applicable	Open	JPG	
20061211_11-06-47-966_NCL200612A_fig 14.jpg	caNanoLab Data	GC000001		Not applicable	Open	JPG	
20061211_11-29-07-542_NCL200612A_fig 27.jpg	caNanoLab Data	GC000001		Not applicable	Open	JPG	
20061211_12-47-57-977_NCL200612A_fig 22.jpg	caNanoLab Data	GC000001		Not applicable	Open	JPG	
20061211_14-49-01-102_NCL200612A_fig 4.jpg	caNanoLab Data	GC000001		Not applicable	Open	JPG	
20061211_14-54-29-189_NCL200612A_fig 5.jpg	caNanoLab Data	GC000001		Not applicable	Open	JPG	
20061211_15-15-15-306_NCL200612A_fig 6.jpg	caNanoLab Data	GC000001		Not applicable	Open	JPG	
20061211_15-25-24-13_N	caNanoLab Data	GC000001		Not applicable	Open	JPG	

Figure 6. Cart page showing options to export or download selected files.

You will be able to download the files or use them to run analysis on the NCI's [Cancer Genomics Cloud](#) (CGC by Velsera) by exporting the list to the CGC first.

There are two ways of exporting the list to CGC:

Option 1: Download Manifest

Click **Download Manifest** to download a manifest file to your computer. The manifest contains:

- DRS URI (File ID)
- File name
- Participant ID
- Md5sumUser Comments

The downloaded manifest when uploaded to your project on the NCI's CGC, will get the files from GC. See next section or [CGC-Page](#) for further details.

Option 2: Export to Cancer Genomics Cloud (Velsera / SBG-CGC)

Click **Available Export Options** → **Export to Cancer Genomics Cloud** to export DRS-based file references to the Velsera Cancer Genomics Cloud platform for cloud-based access and analysis.

You will be redirected to [Cancer Genomics Cloud](#) (Figure 7). You must log in or create an account to complete the export.

After authentication:

- 1) Select or create a destination project
- 2) Confirm the project selection
- 3) Agree to the CGC terms and Data Use Agreement
- 4) Complete the import process

The selected files will then be accessible within your CGC project workspace for downstream

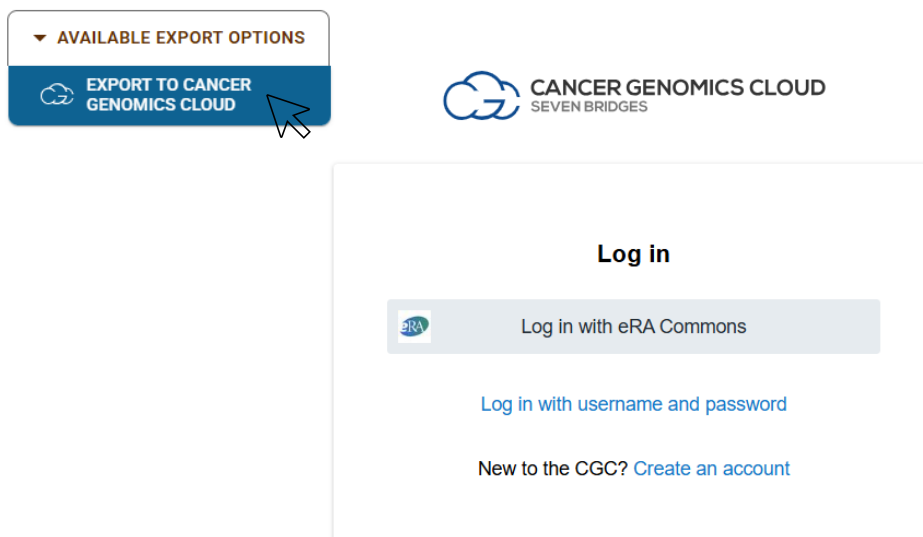


Figure 7. The Available Export Options menu displaying the “Export to Cancer Genomics Cloud” option and the resulting redirect to the Cancer Genomics Cloud login page.

Cart Page Help (README)

The Cart page includes a README button (Figure 8) that provides detailed background information about:

- The CRDC ecosystem
- The purpose and structure of file manifests
- DRS identifiers
- Integration with the Cancer Genomics Cloud (CGC)

Users seeking additional technical details on file access and cloud-based workflows may refer to the README pop-up directly from the Cart page. The content can also be downloaded as a PDF.

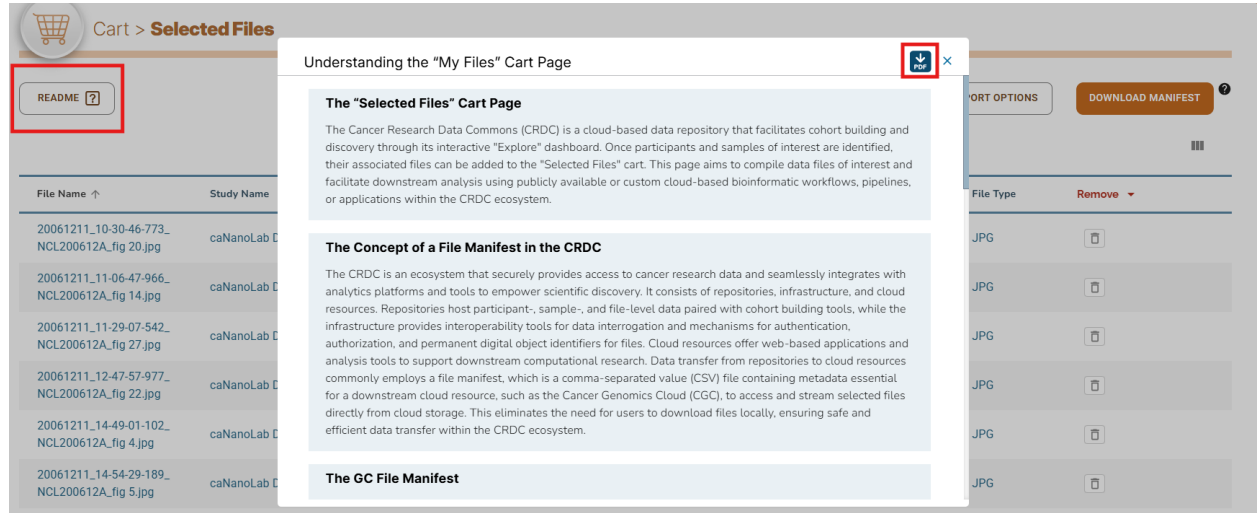


Figure 8. README pop-up window providing detailed information about file manifests and CGC integration.

Section 2: Submitting Data in General Commons (GC)

I. Introduction

The NCI Cancer Research Data Commons (CRDC) is a secure, cloud-based infrastructure that supports the submission, sharing, and analysis of cancer research data. Through its integrated

Data Commons, CRDC enables standardized data management, interoperability, and controlled access across NCI-funded programs.

caNanoLab data has now been integrated into the CRDC's General Commons (GC) . This integration allows Nanotechnology data, including Samples, Protocols, Publications, Characterizations, Composition, and related Files to be submitted, managed, and accessed within the GC data model.

This guide provides step-by-step instructions for

- Submit a Submission Request Form through the [CRDC Submission Portal](#).
- requesting a Submitter role
- creating and managing submissions
- downloading and completing submission templates, and
- uploading, validating, and finalizing caNanoLab data submissions in GC

II. Prerequisites

Before initiating a data submission to General Commons (GC), ensure the following:

- **CRDC Portal Account:** Create a [Login.gov](#) account to access the CRDC Submission Portal. NIH staff may log in using a PIV card.
- **Submission Approval:** Submit a Submission Request Form through the [CRDC Submission Portal](#). Detailed instructions can be found [here](#)).
- **Submitter Role:** You must request and be granted the Submitter role in the CRDC Submission Portal before submitting data.

III. Requesting the Submitter Role

By default, new accounts on CRDC Submission Portal are assigned a User role, which does not allow data submission. The Submitter role must be requested and approved before you can begin.

Steps to Request the Submitter Role

1. Log in to the CRDC Submission Portal.

Click your name in the upper-right corner of the page to access your **User Profile** (See Figure 1).

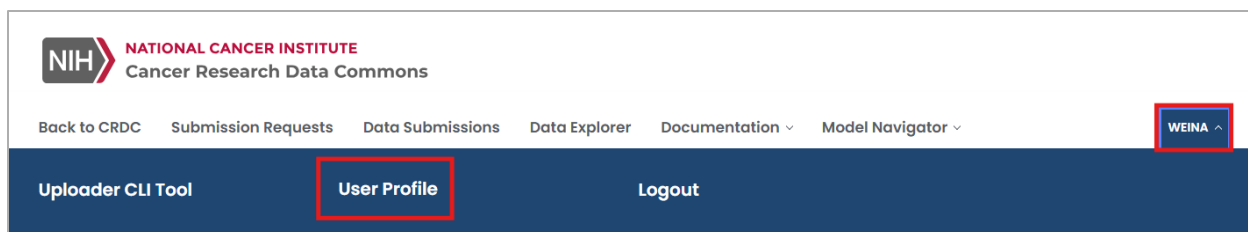


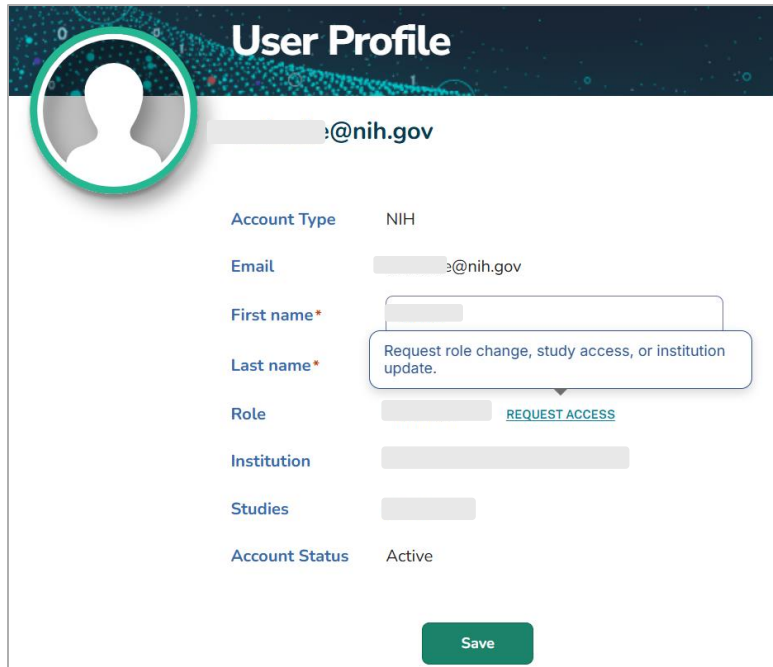
Figure 1. Access to User Profile.

2. Click “Request Access.”

The Request Access button allows users to:

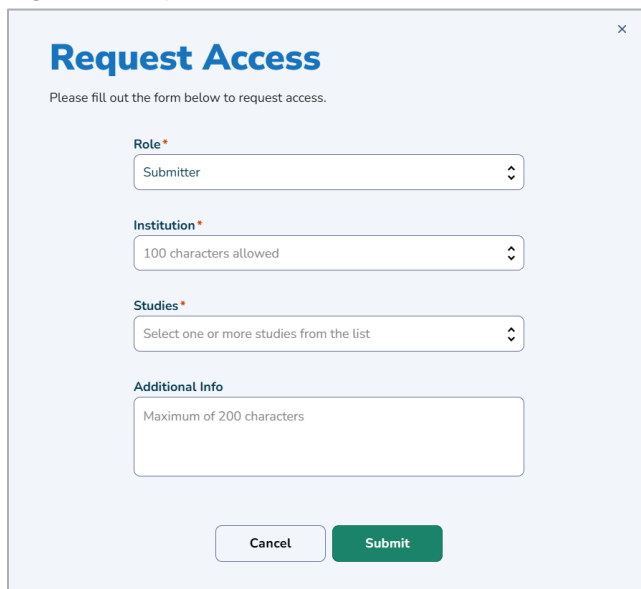
- Request the Submitter role (required for data submission),
- Request access to specific studies, or
- Update the institution associated with their account.

See Figure 2 and Figure 3.



The image shows a 'User Profile' page. At the top, there is a header with a user profile icon and the text '@nih.gov'. Below this, the profile details are listed: Account Type (NIH), Email (@nih.gov), First name, Last name, Role, Institution, Studies, and Account Status (Active). A green 'REQUEST ACCESS' button is located next to the Role field. A tooltip is visible over the button, stating: 'Request role change, study access, or institution update.'

Figure 2. Request access to a Submitter role.



The image shows a 'Request Access' modal form. It has a title bar with a close button (X). The form contains the following fields: Role (dropdown menu with 'Submitter' selected), Institution (dropdown menu with '100 characters allowed' selected), Studies (dropdown menu with 'Select one or more studies from the list' selected), and Additional Info (text area with 'Maximum of 200 characters' placeholder). At the bottom, there are 'Cancel' and 'Submit' buttons.

Figure 3. Request access form.

3. Complete the Request Access Form.

In the form:

- Select Submitter from the Role dropdown menu.
- Select your institution from the Institution dropdown menu.
- If your institution is not listed, enter it manually.
- Select the relevant study or studies from the Studies dropdown menu.
- (Optional) Provide additional context in the Additional Information field to clarify your role or responsibilities.

4. Submit the request.

The CRDC System Administrator will review your request. Once approved, you will be notified through your login identity. After approval, you will be able to initiate and manage data submissions in the portal.

Note: Accounts that are inactive for 60 days may be automatically deactivated. If this occurs, contact the CRDC Help Desk (NCICRDC@mail.nih.gov) to restore access.

IV. Starting a New Submission

You can start a new submission once you finish the prerequisites listed above.

Log in with a Submitter role, navigate to the Data Submissions tab on the CRDC Submission portal (Figure 4).

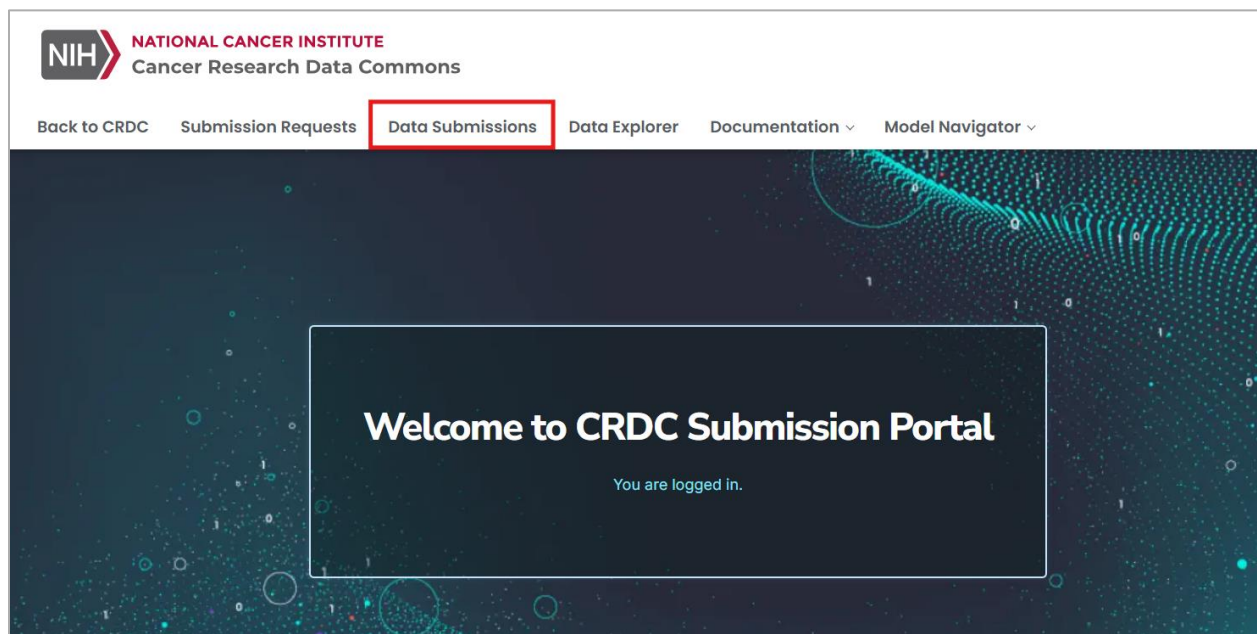


Figure 4. Data Submission Menu to start a submission.

If this is your first submission, the data submission list will appear empty. Otherwise, all existing submissions will be displayed on the List page. See Figure 5.

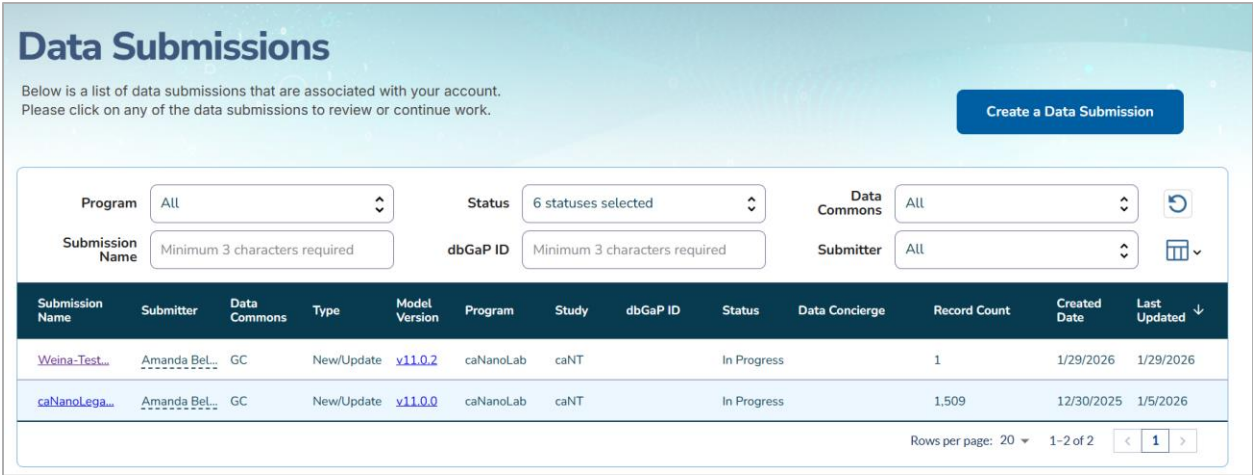


Figure 5. Data Submission Home.

To initiate a new data submission, click the **Create a Data Submission** button as shown in Figure 6. A dialog box will appear to begin the submission.

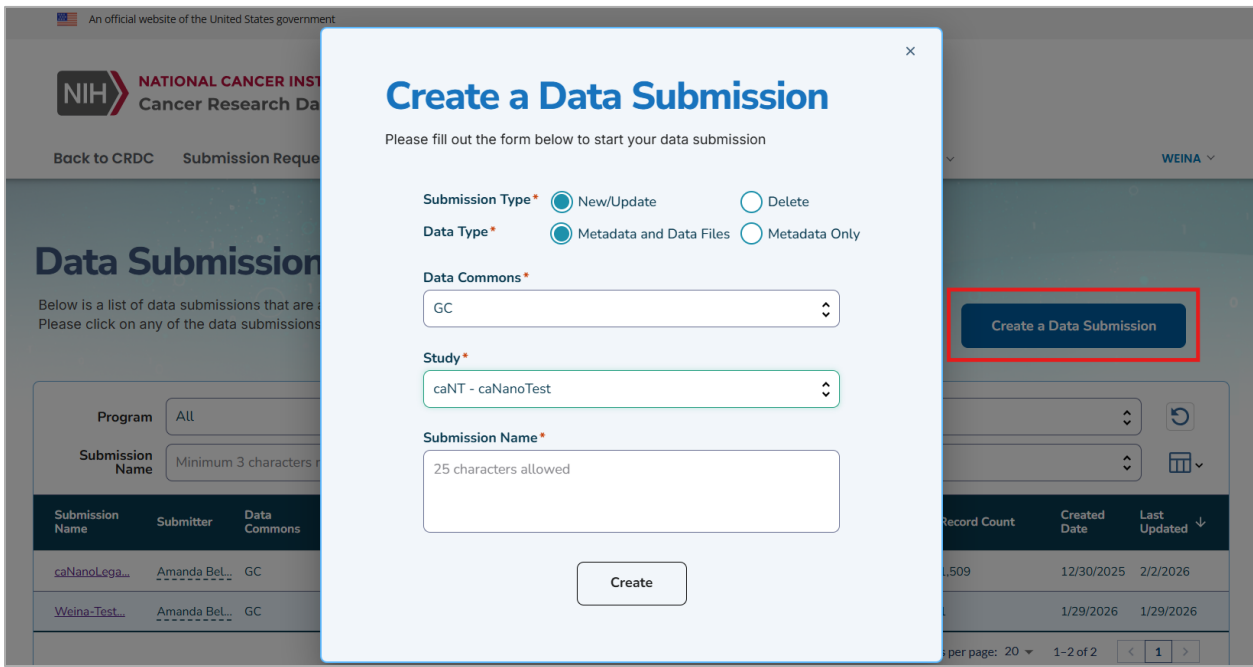


Figure 6. Data Submission Dialog Box.

Complete all required fields in the dialog box:

1. **Submission Type:** Choose the appropriate action:

- Select New/Update to start a new submission or update an existing one.
- Select Delete to request the removal of publicly released files.

Note: For deletion, submit a Metadata Only file. Once approved, the CRDC team will process the removal.

2. **Data Type:** Select one of the following:

- Metadata and Data Files – if submitting both.
- Metadata Only – if submitting only metadata.

3. **Data Commons:** For Nanotechnology submissions, select “GC.” If “GC” does not appear in the list, email the CRDC Help Desk at NCICRDC@mail.nih.gov.

4. **Study:** Choose from the dropdown menu showing the Study Title or Study Abbreviation you previously submitted (e.g., “caNT-caNanoTest”).

If you notice an error, contact the CRDC Help Desk at NCICRDC@mail.nih.gov.

5. **Submission Name:** Enter a descriptive name in the free-text field to label your submission. This name will appear in the Data Submission List once the submission is created. Use a meaningful and informative name to make future tracking easier.

6. Click **Create:** This button becomes active only after all required fields are correctly completed.

Once the submission is created, CRDC assigns a Data Concierge to your submission.

You can find their email address on your data submission dashboard.

From this point onward, direct all submission-related questions to your assigned Data Concierge.

V. Continuing an Existing Submission

1. **Navigate to the “Data Submissions” tab.**

You’ll see a table listing all your submissions, including any newly created ones.

2. **Locate your submission.**

Find the submission you wish to update under the “**Submission Name**” column.

- To view the full name (if it’s truncated), hover your cursor over the name (see Figure 7).

3. **Customize the table view (optional).**

- Click the Table icon in the top-right corner to choose which columns to display.
- Select your desired columns and apply the changes.

4. **Use filters to refine the list (optional).**

- When there are many submissions, use the filter options at the top of the table to narrow down your search.

NIH NATIONAL CANCER INSTITUTE
Cancer Research Data Commons

Back to CRDC Submission Requests **Data Submissions** Data Explorer Documentation Model Navigator WEINA

Data Submissions

Below is a list of data submissions that are associated with your account. Please click on any of the data submissions to review or continue work.

Create a Data Submission

Program: All Status: 6 statuses selected Data Commons: All Submission Name: Minimum 3 characters required dbGaP ID: Minimum 3 characters required Submitter: All

Submission Name	Submitter	Data Commons	Type	Model Version	Program	Study	dbGaP ID	Status	Data Concierge	Record Count	Created Date	Last Updated
caNanoLega...	Amanda Bel...	GC	New/Update	v11.0.0	caNanoLab	caNT		In Progress		1,509	12/30/2025	2/2/2026
Weina-Test...	Amanda Bel...	GC	New/Update	v11.0.2	caNanoLab	caNT		In Progress		1	1/29/2026	1/29/2026

Rows per page: 20 1-2 of 2

Figure 7. Active Data Submissions List.

5. Open an Existing Submission.

To continue working on a submission, click the relevant **Submission Name**. This will take you to the data upload interface.

6. Upload data.

To accurately associate your uploaded data with the corresponding nodes and ensure a successful upload, you must use the Submission Templates. After downloading the required templates, refer to Section VII for uploading instructions.

VI. Obtaining Submission Templates and Reviewing Node Properties

To successfully upload metadata files and accurately associate your data with the correct nodes in the General Commons (GC) model used by caNanoLab, follow these steps to obtain the required submission templates and review the data elements they contain:

1. Navigate to the GC Model Viewer

- Click on the **Model Navigator** tab in the top menu bar (see Figure 8).
- Select the GC Model. You will be redirected to the GC Data Model Viewer page.

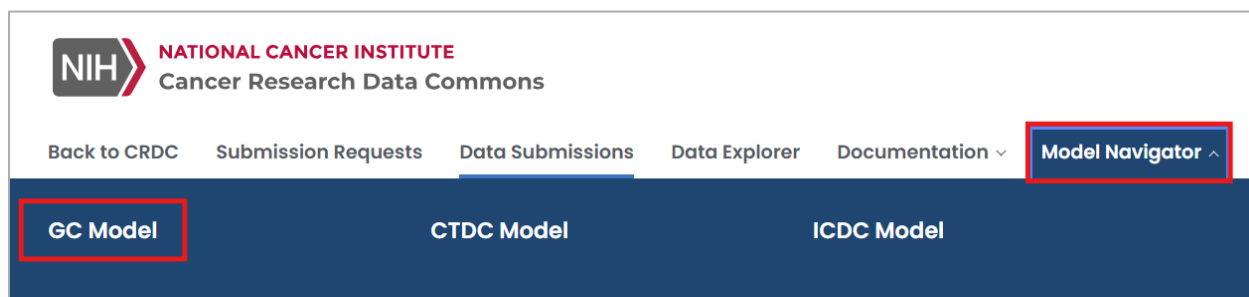


Figure 8. Use the Menu Bar to Navigate to the Data Model Viewer.

2. Download Required Templates

- In the GC Data Model Viewer page, locate the **Available Downloads** dropdown menu under the Graph View (see Figure 9).
- Select **Submission Template > Selective Template**.
- Select and download the following templates:
 - ✓ Program
 - ✓ Study
 - ✓ Investigator
 - ✓ Sample
 - ✓ Composition
 - ✓ Characterization
 - ✓ Publication
 - ✓ Protocol
 - ✓ File
- Click **Download** after all the sections are selected (see Figure 10).

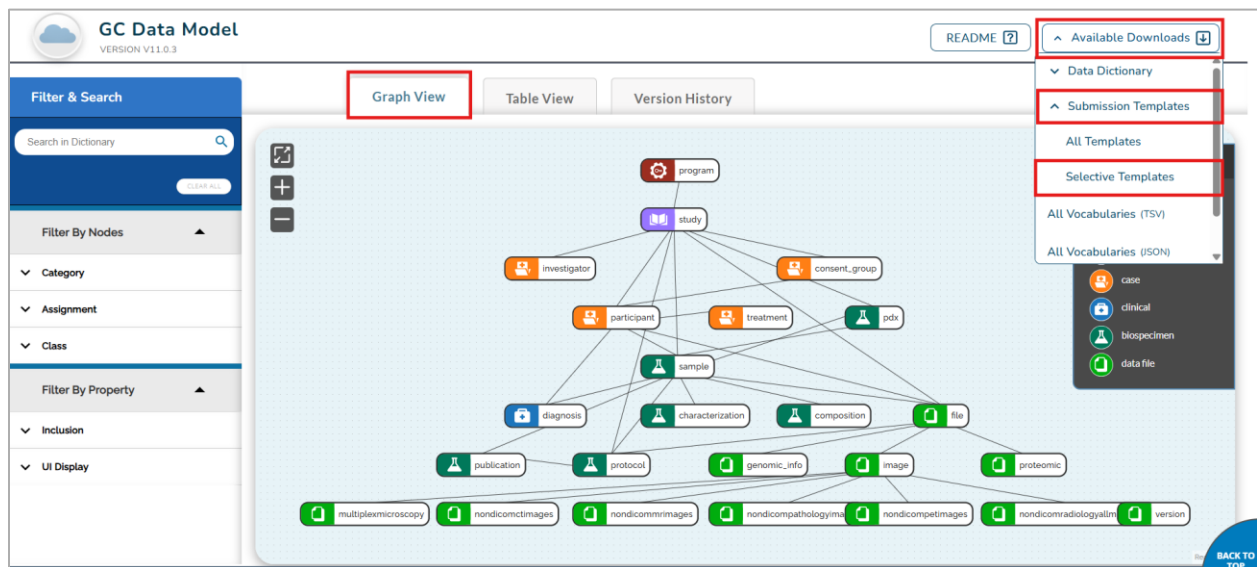


Figure 9. Using the Available Downloads Menu under Graph View.

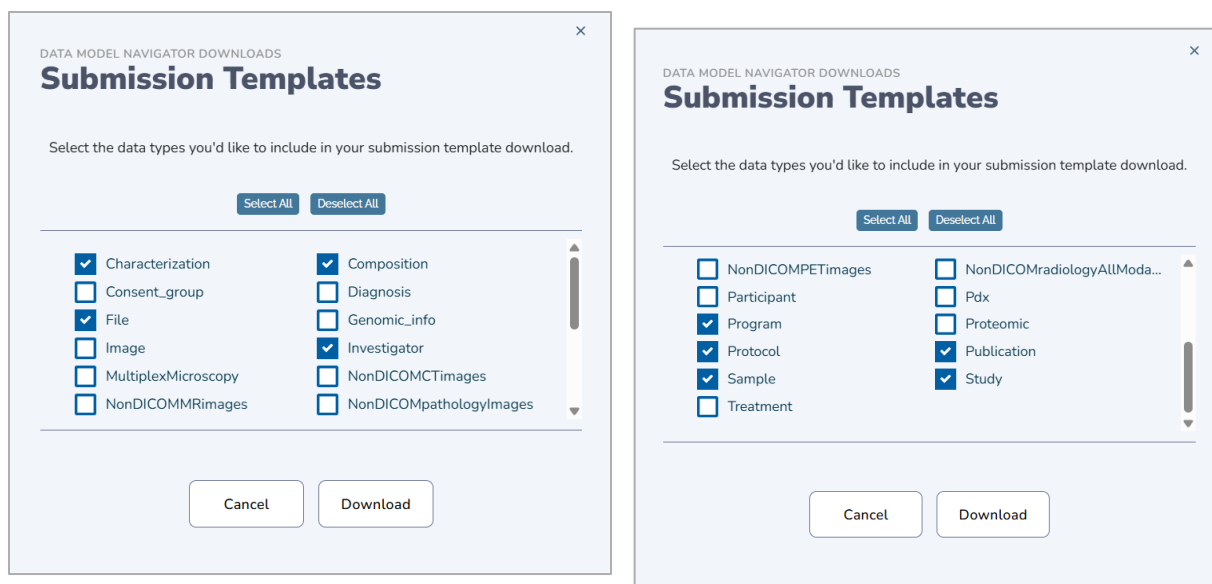


Figure 10. Download Selected Submission Templates.

3. Review Data Elements

- In the **Graph View** tab, the data model is displayed as graphical nodes and relationships (e.g., Sample, Characterization).
- Click on a node to open a pop-up dialog box.
- In the dialog, click View Properties to access a Table View of that node's data elements (see Figure 11).

Alternatively, properties of each node can be reviewed under Table View. The Table View lists all the data elements/properties of the data model.

- You can go directly to the **Table View** tab (see Figure 12).
- Click the **number of Properties** to view its full list of properties.
- Review the description of each property, such as data type, whether it is required or optional for submission, etc.
- These properties correspond to columns in the Excel templates.
- Ensure all required fields are accurately completed in your metadata files.

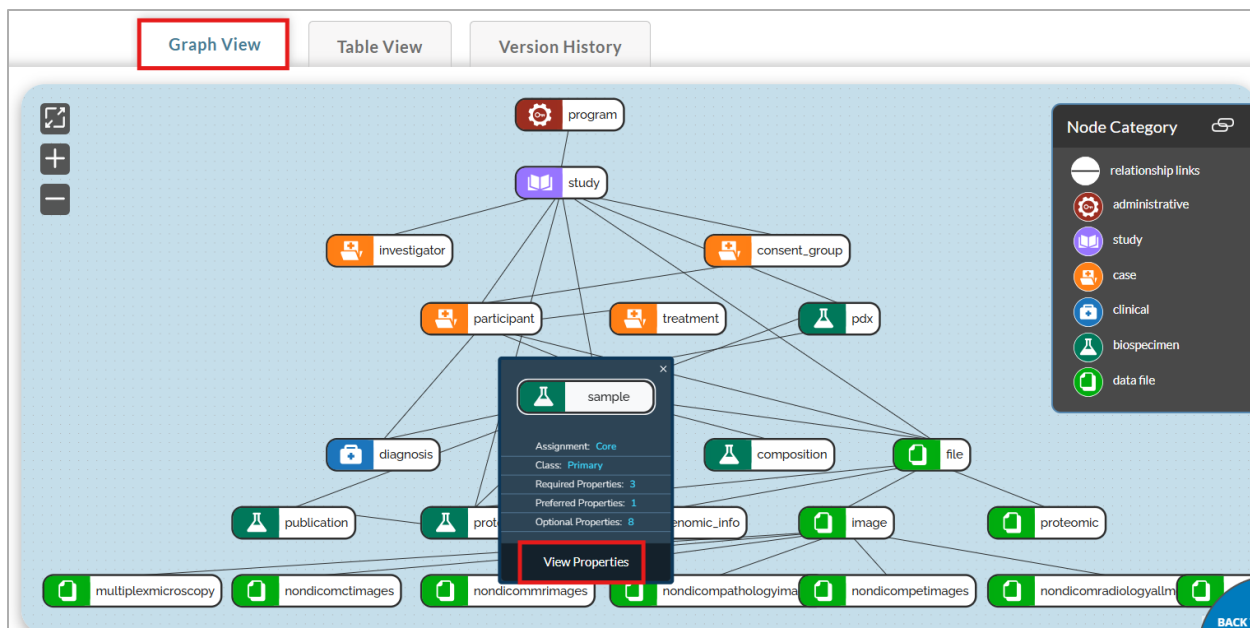


Figure 11. Click a Node to View its properties using Graph View.

Graph View **Table View** Version History

21 Properties 2 Relationships Template Data Dictionary

Biospecimen

Sample Specimen tissue type collected from the participant. Assignment: Core Class: Primary

12 Properties 3 Relationships Template Data Dictionary

Property	Type	CDE Info	Required	Description	Source
sample_id	"string"	CDE Full Name Biospecimen Source Laboratory Identifier Version 1.00 Public ID 6921892 Origin caDSR	Required	Sample identifier as submitted by requestor	
sample_type	Acceptable Values: • Analyte	CDE Full Name Specimen Material ORIR	Required	Tissue type of this sample	

BACK TO

Figure 12. Table View of an example of the Sample Node.

VII. Uploading and Submitting Metadata

The CRDC Submission Portal offers a **Uploader CLI Tool** for uploading data to temporary CRDC storage. The behavior of the Uploader CLI Tool is managed by a **Configuration File**. For CLI Uploader Tool and Configuration File installation/downloading and usage instructions, refer to the [CRDC Data Submission Instructions](#).

1. Uploading Metadata Files

- Click the **Submission Name** you want to work with to submit data (see Figure 7). A new submission interface will pop up (see Figure 13).
- Click **Choose Files** and select the filled Excel templates, the number of files selected will be indicated.
- Click **Upload** to submit the file(s) you just selected (See Figure 14).
- Monitor the **Status** column for upload confirmation.
 - ✓ If a file fails, it will be marked as "Failed". To remove failed files, go to the Data View table and click the Delete icon.

The screenshot displays the CRDC Submission Portal interface for a submission named "Weina-Test-Tuto...". The submission ID is 3b33f4ed-110d-4e51-a37e-082a1cea3530. The submission type is "New/Update", the submitter is "Amanda Bell", the study is "caNT", and the program is "CANL". The data commons is "GC" and the data concierge is "DATA CONCIERGE".

A large white box with a blurred background image of a bar chart and a pie chart contains the text: "This is the data submission visualization section which displays validation results for uploaded data. After uploading and validating the data (see below), the visualization graphic will display."

The main interface is divided into three steps:

- 1. UPLOAD METADATA** (GC DATA MODEL: v11.0.2)
 - Metadata Files: Choose Files (No files selected)
 - Upload button
- 2. UPLOAD DATA FILES** (UPLOADER CLI VERSION: v4.3)
 - Description: The CLI Tool is used to upload data files to CRDC Submission Portal and requires a configuration file to work. The CLI Tools is a one-time download however the configuration file needs to be customized for each submission. You can either edit the example configuration file found in the [CLI Tool download](#), or you can click the button on the right to download a configuration file customized for this submission.
 - Download Configuration File button
- 3. VALIDATE DATA**
 - Validation Type: ☒ Validate Metadata, ☐ Validate Data Files, ☐ Both
 - Validation Target: ☒ New Uploaded Data, ☐ All Uploaded Data
 - Validate button

At the bottom, there are tabs for "Upload Activities", "Validation Results", and "Data View". The "Data View" tab is active, showing a table with the following columns: Batch ID, Batch Type, File Count, Status, Uploaded Date, Uploaded By, and Upload Errors.

Batch ID	Batch Type	File Count	Status	Uploaded Date	Uploaded By	Upload Errors
----------	------------	------------	--------	---------------	-------------	---------------

Figure 13. Submission interface.

The screenshot shows the submission interface with a top navigation bar and a main content area. The top bar includes buttons for 'Review Comments' and 'Full History', and displays submission details: SUBMITTER (Amanda Bell), STUDY (caNT), PROGRAM (CANL), COLLABORATORS (0), DATA COMMONS (GC), and DATA CONCIERGE. The main content area has a large placeholder for data submission visualization. Below this, there are three main sections: 1. UPLOAD METADATA (GC DATA MODEL: v11.0.2) with a 'Metadata Files' section showing '1 file selected' and an 'Upload' button highlighted with a red box. 2. UPLOAD DATA FILES (UPLOADER CLI VERSION: v4.2) with instructions and a 'Download Configuration File' button. 3. VALIDATE DATA with radio buttons for 'Validation Type' (Validate Metadata, Validate Data Files, Both) and 'Validation Target' (New Uploaded Data, All Uploaded Data), and a 'Validate' button. At the bottom, there are tabs for 'Upload Activities', 'Validation Results', and 'Data View', and a table header for 'Batch ID', 'Batch Type', 'File Count', 'Status', 'Uploaded Date', 'Uploaded By', and 'Upload Errors'.

Figure 14. One file is selected to be uploaded.

2. Validating Metadata

- Validations can be run at any point in the submission process,
- Select options in the Validate Data panel:
 - ✓ Validation Type: The **Validate Metadata** option runs validations only on the submitted Metadata Manifest, not on any of the uploaded data files. The **Validate Data Files** option does the reverse and checks all the uploaded data files. The **Both** option validates both.
 - ✓ Validation Target: By default, only **New Uploaded Data** is validated. If there is a need to check the entire submission, the **All Uploaded Data** option validates everything that has been uploaded so far.
- Click on **Validate** (see Figure 15).

3. Submitting the Final Dataset

- Once validations pass, click **Submit** (see Figure 15).

- If submitted in error, contact your Data Concierge.

Prior to beginning uploading process, read detailed instructions available in the [Data Submission Instructions](#).

1

UPLOAD METADATA

GC DATA MODEL: [v11.0.2](#)

Metadata Files

Choose Files

No files selected

Upload

2

UPLOAD DATA FILES

UPLOADER CLI VERSION: [v4.3](#)

The CLI Tool is used to upload data files to CRDC Submission Portal and requires a configuration file to work. The CLI Tools is a one-time download however the configuration file needs to be customized for each submission. You can either edit the example configuration files found in the [CLI Tool download](#), or you can click the button on the right to download a configuration file customized for this submission.

Download Configuration File

3

VALIDATE DATA

Validation Type:

☒ Validate Metadata
 ☐ Validate Data Files
 ☐ Both

Validation Target:

☐ New Uploaded Data
 ☒ All Uploaded Data

Validate

Upload Activities

Validation Results

Data View

dbGaP Sheets

Batch ID	Batch Type	File Count	Status	Uploaded Date ↓	Uploaded By	Upload Errors
1	Metadata	1	Uploaded	1/29/2026	Amanda Bell	

Rows per page: 20

1-1 of 1

< 1 >

← Back to Data Submissions

Submit

Cancel

Figure 15. After all submissions are validated, you can officially submit your data.

4. After Submission

- The CRDC Submission Team will review your submission.
- If everything passes CRDC validation, you'll receive confirmation, and the data will be indexed and released.
- If issues are found, you'll be contacted for revisions, and the submission may be reopened for correction.